

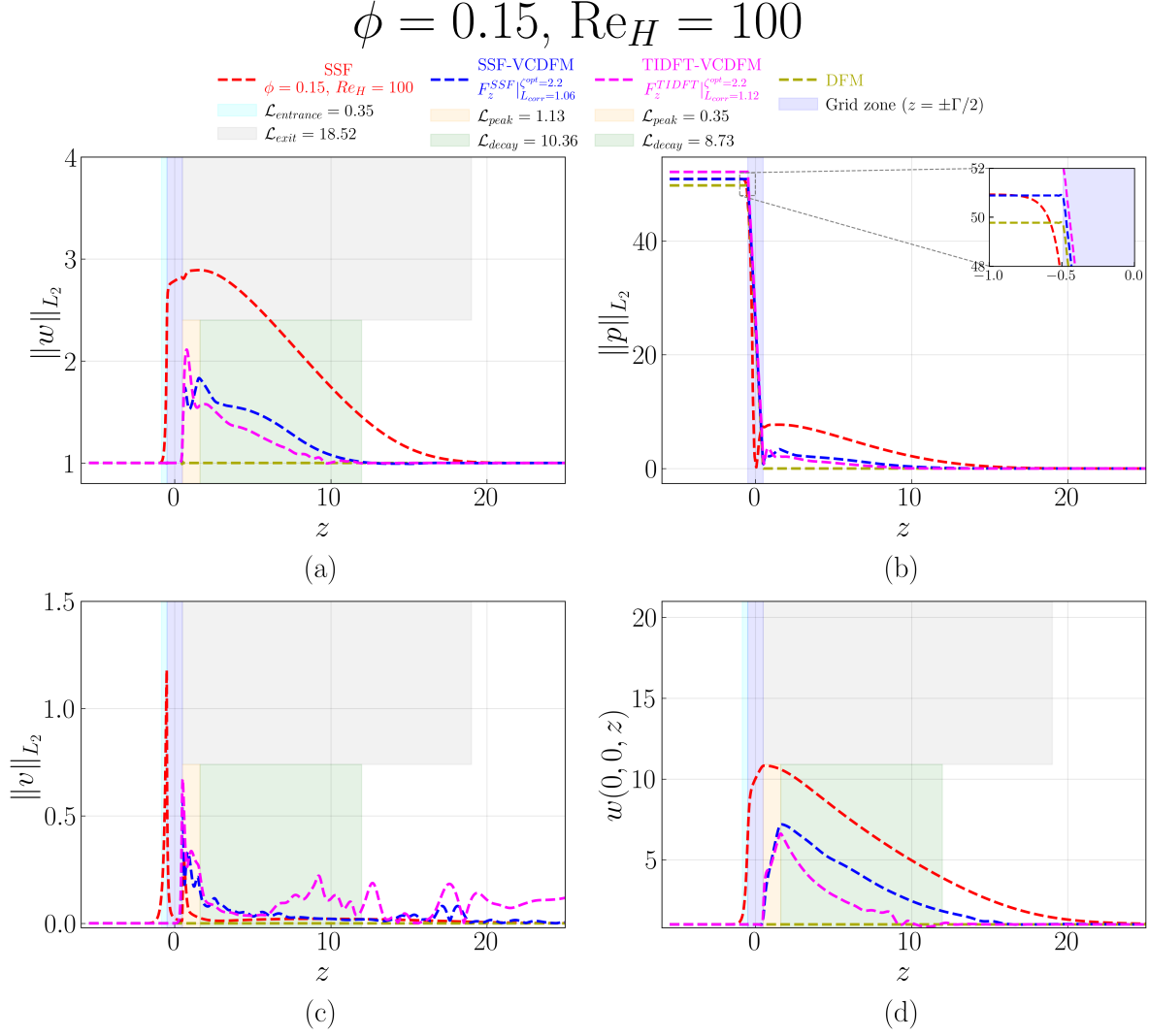


Figure D.241: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.15$ and $\text{Re}_H = 100$. The figure compares the SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM values appear in the legend.

$$\phi = 0.15, \text{Re}_H = 100$$

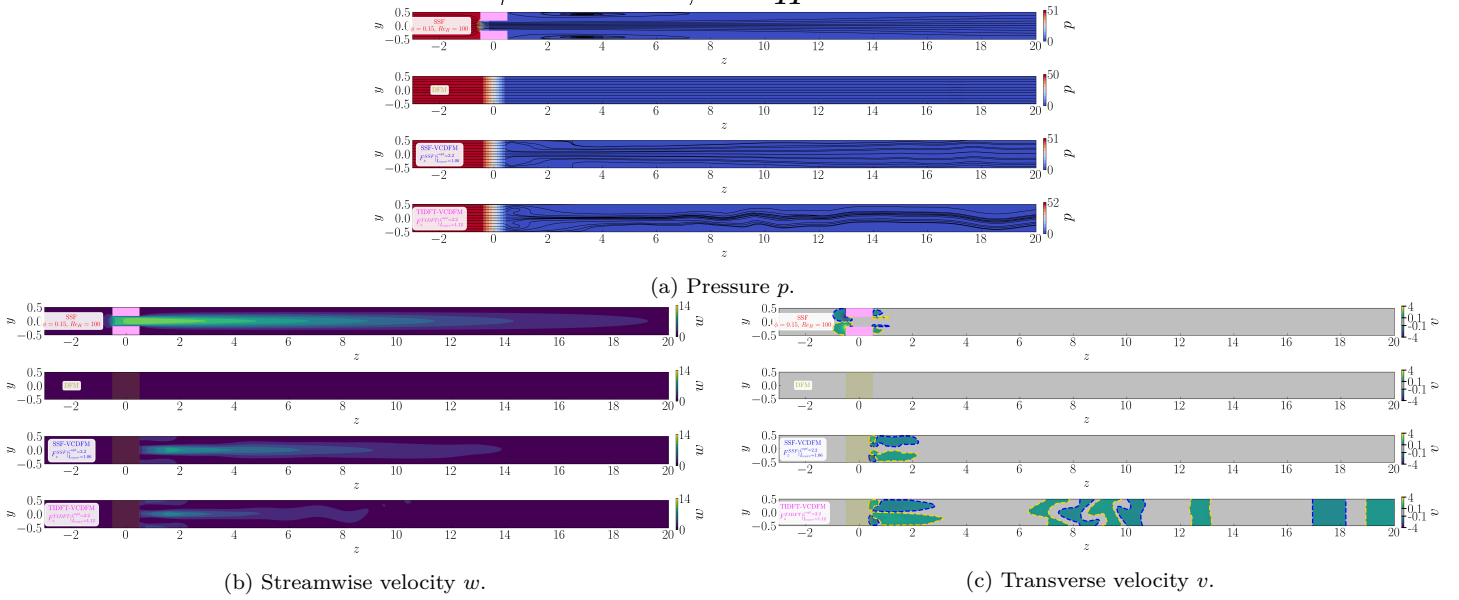


Figure D.242: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.15$, $\text{Re}_H = 100$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

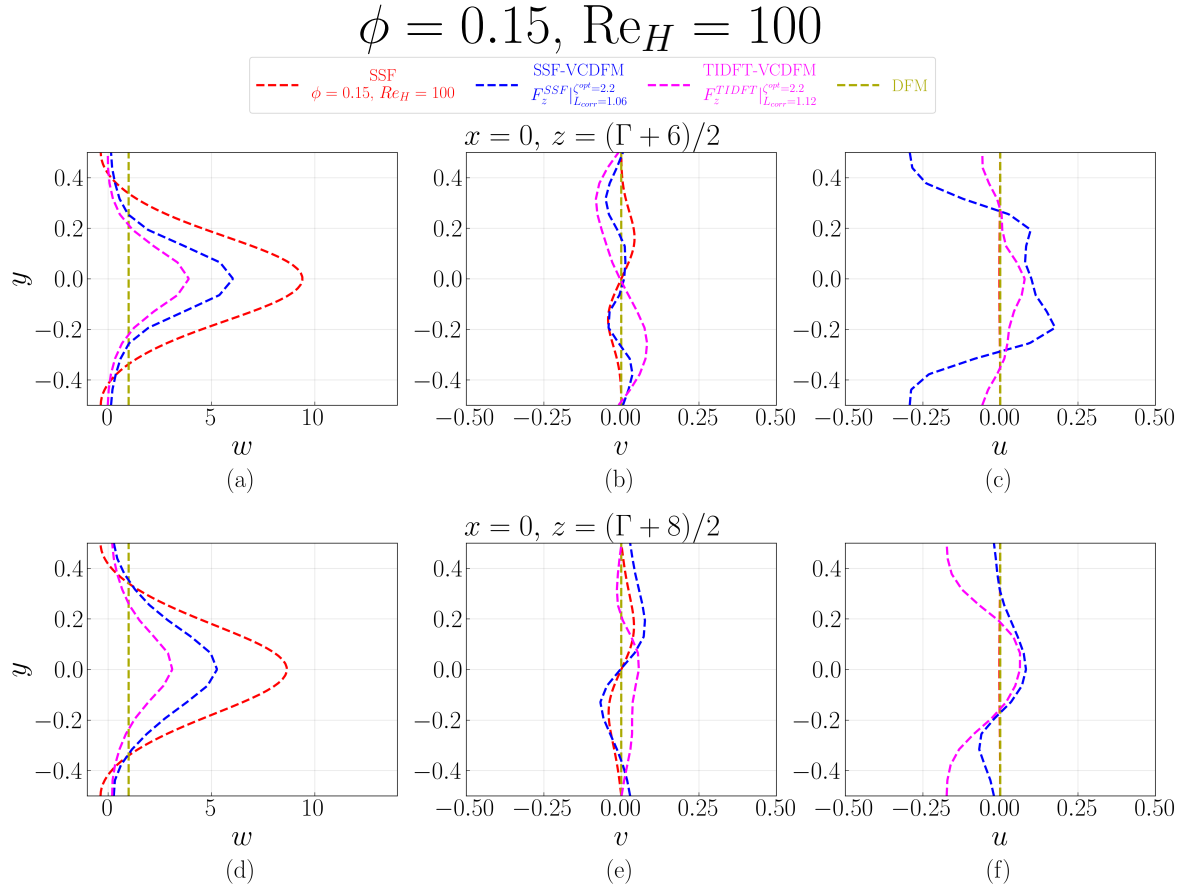


Figure D.243: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.15$ and $\text{Re}_H = 100$, comparing the SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (\Gamma + 6)/2$ in (a–c) and $z = (\Gamma + 8)/2$ in (d–f).

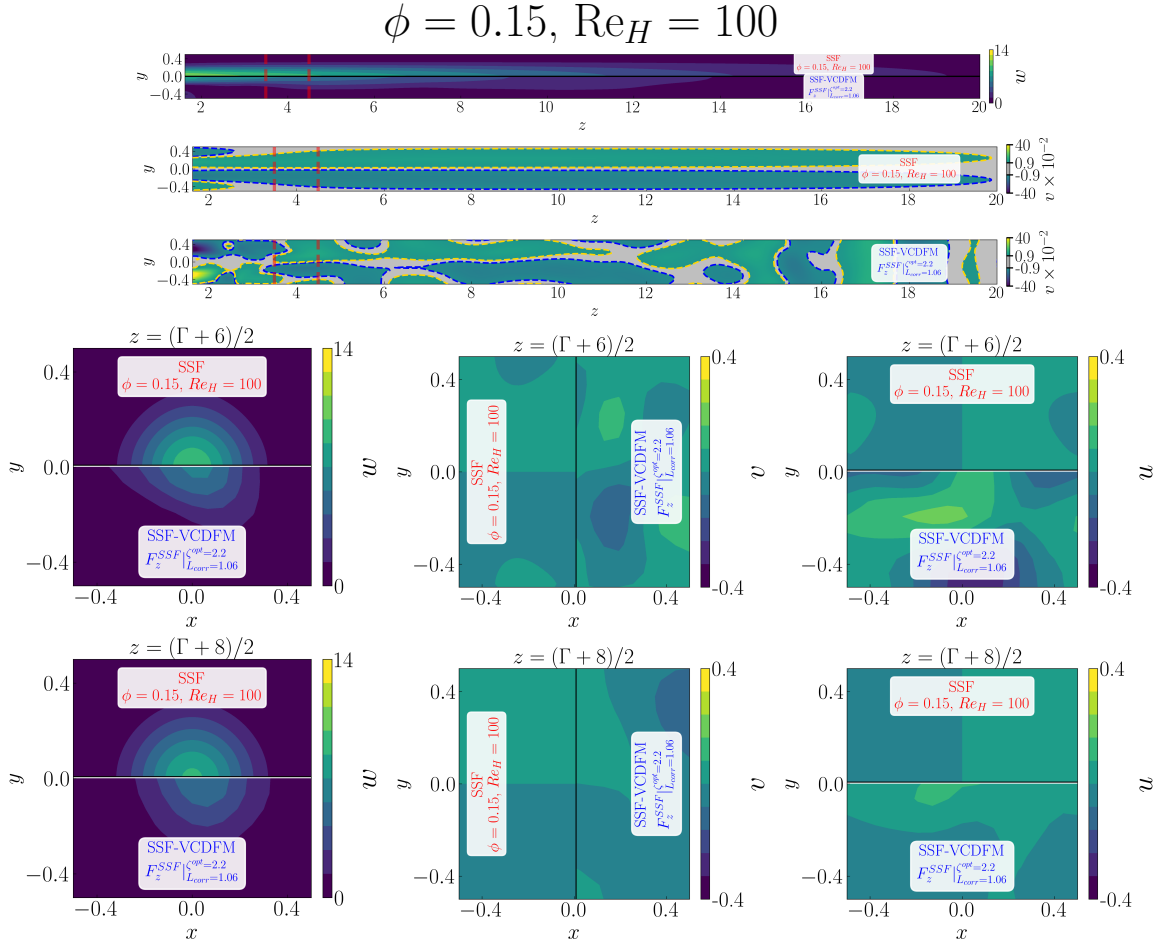


Figure D.244: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.15$ and $Re_H = 100$. The upper panels show the downstream flow in the streamwise (z, y) -plane, restricted to the correction region. The lower panels show the corresponding transverse (x, y) -plane distributions at $z = (\Gamma + 6)/2$ and $z = (\Gamma + 8)/2$.

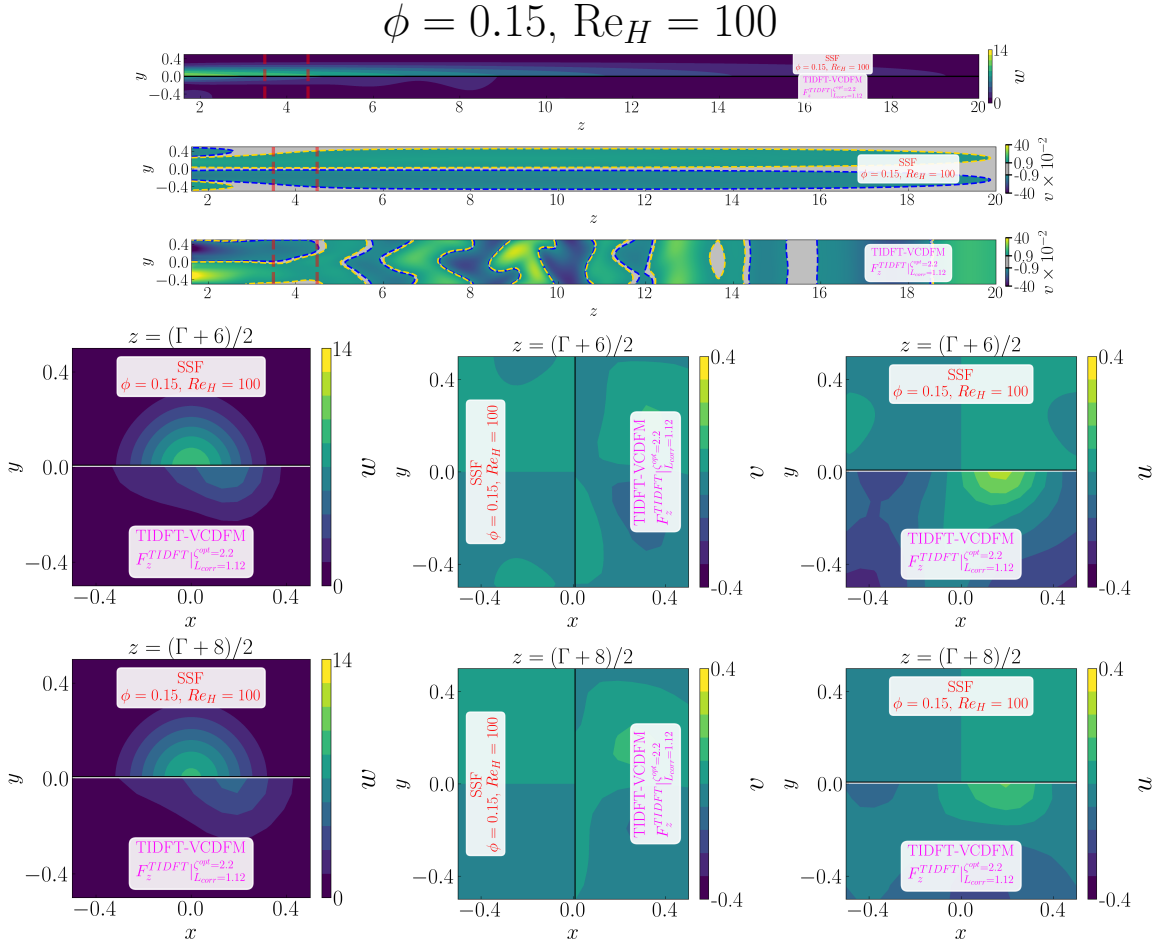


Figure D.245: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.15$ and $\text{Re}_H = 100$. The upper panels compare the downstream flow in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions at $z = (\Gamma + 6)/2$ and $z = (\Gamma + 8)/2$.

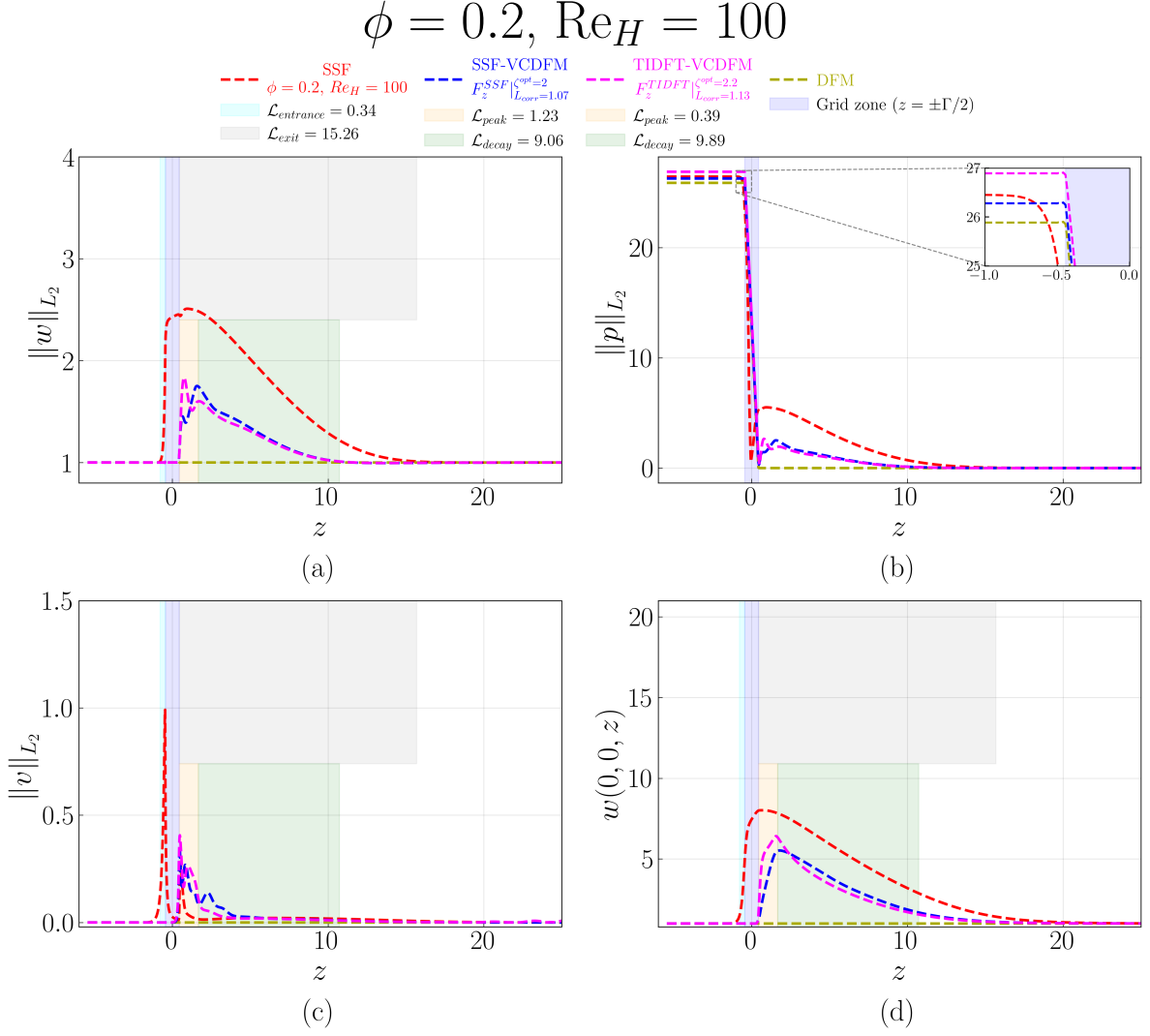


Figure D.246: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.2$ and $\text{Re}_H = 100$. The figure compares the SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM values appear in the legend.

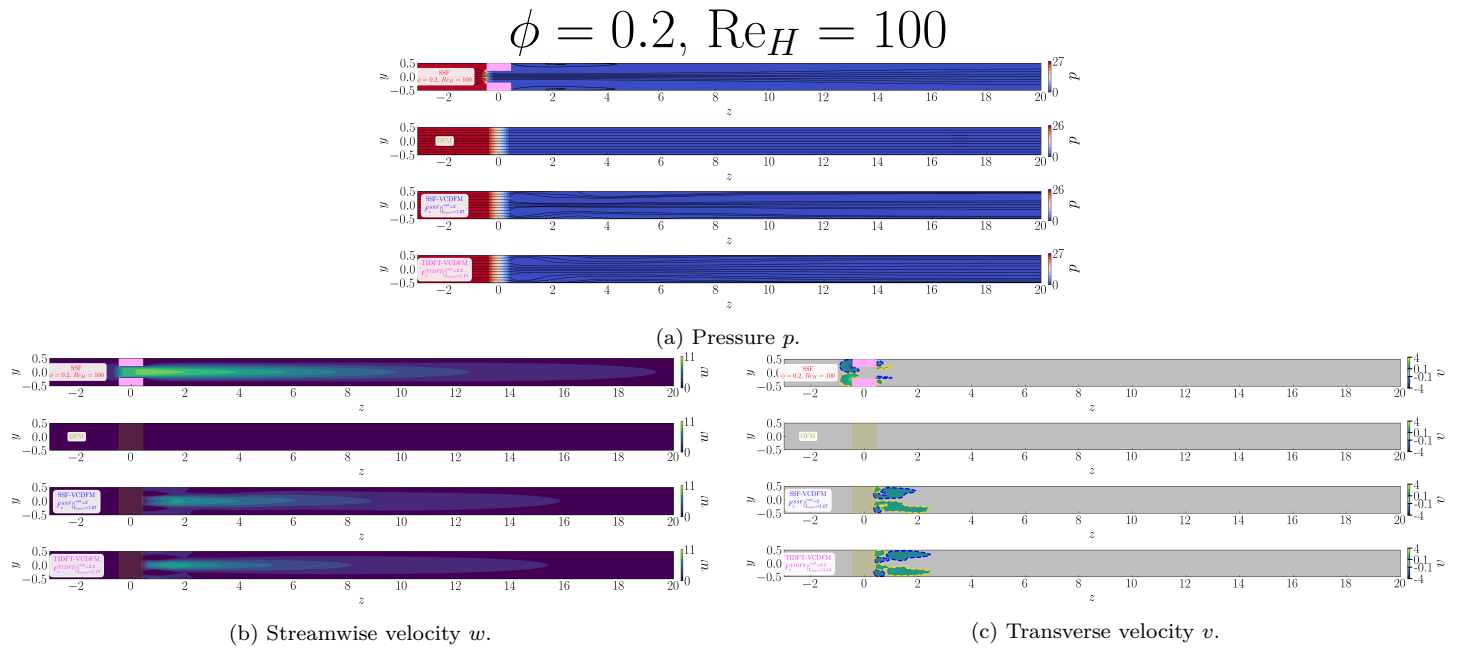


Figure D.247: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.2$, $\text{Re}_H = 100$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

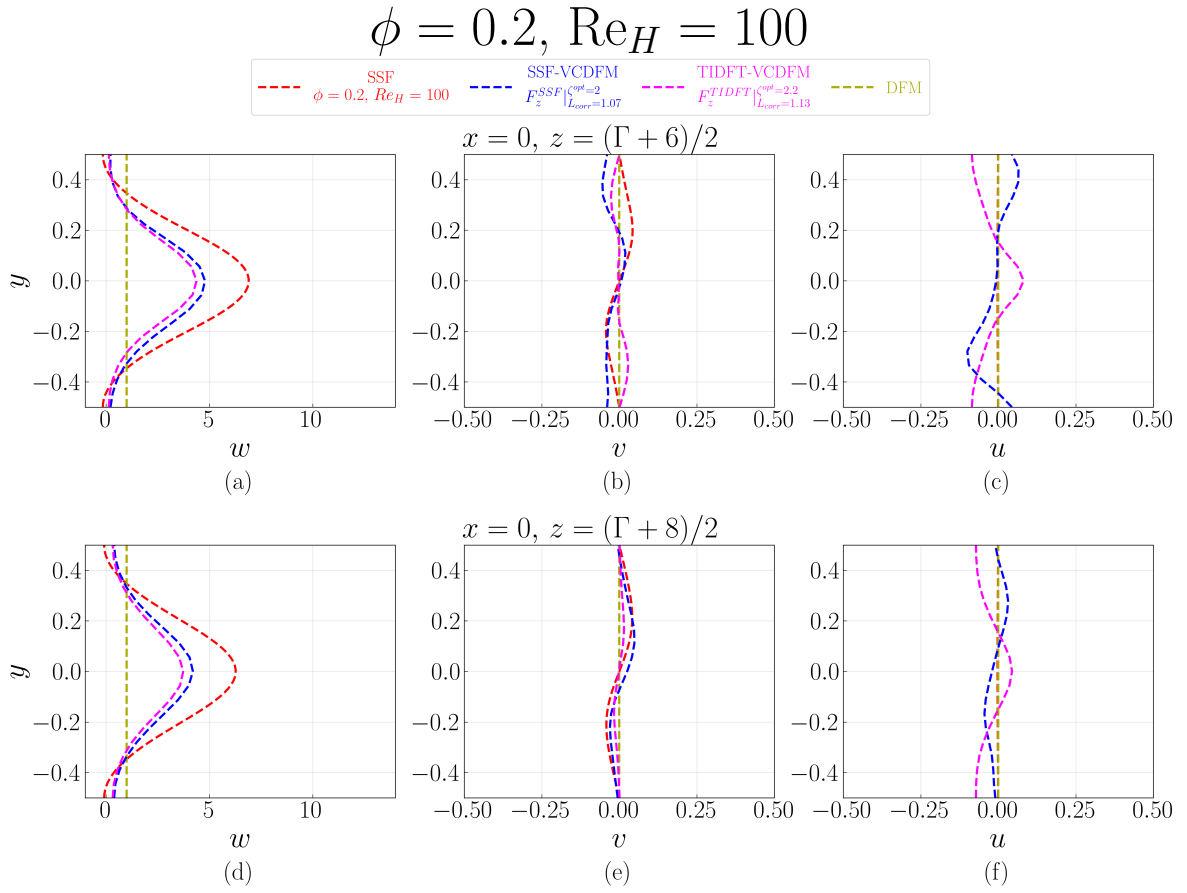


Figure D.248: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.2$ and $\text{Re}_H = 100$, comparing the SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (\Gamma + 6)/2$ in (a-c) and $z = (\Gamma + 8)/2$ in (d-f).

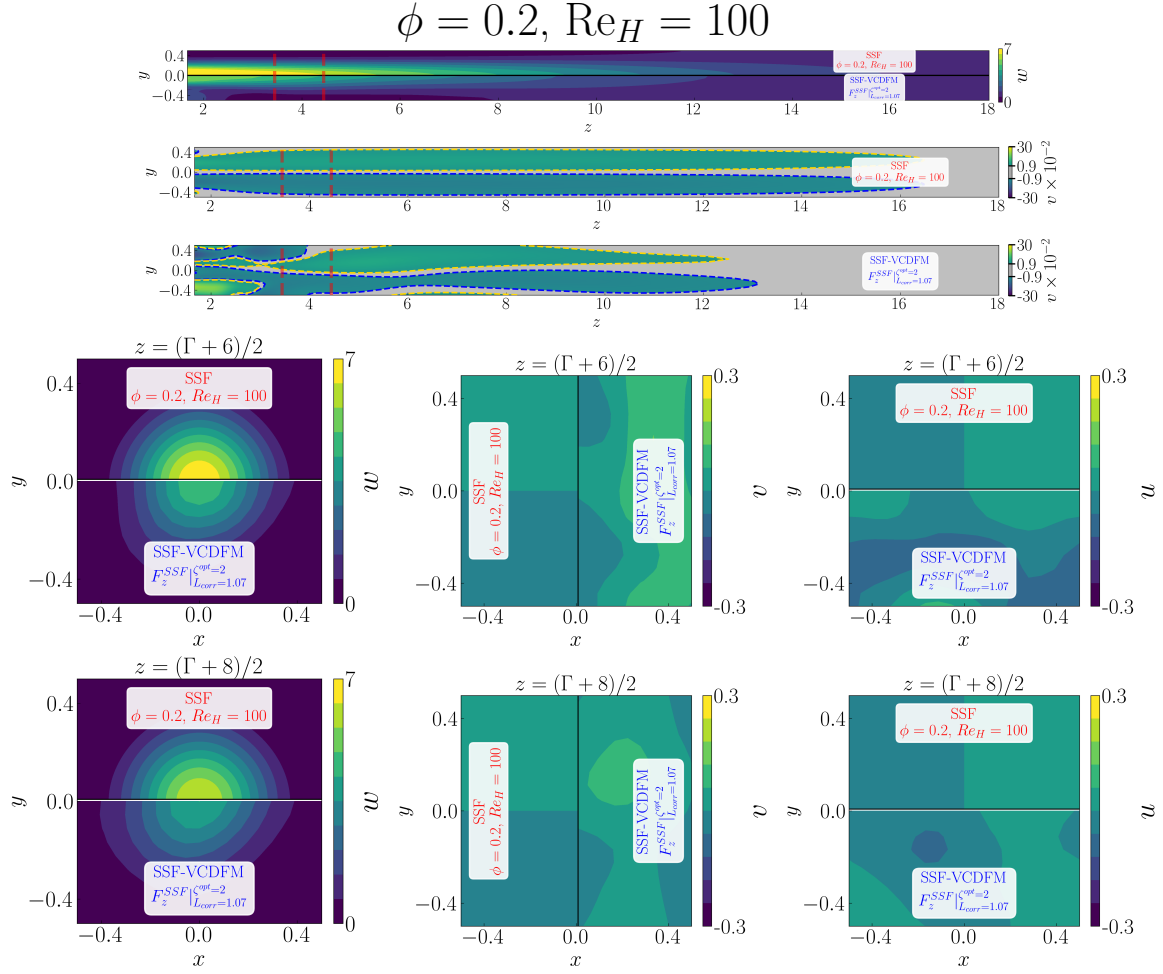


Figure D.249: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.2$ and $\text{Re}_H = 100$. The upper panels show the downstream flow in the streamwise (z, y) -plane, restricted to the correction region. The lower panels show the corresponding transverse (x, y) -plane distributions at $z = (\Gamma + 6)/2$ and $z = (\Gamma + 8)/2$.

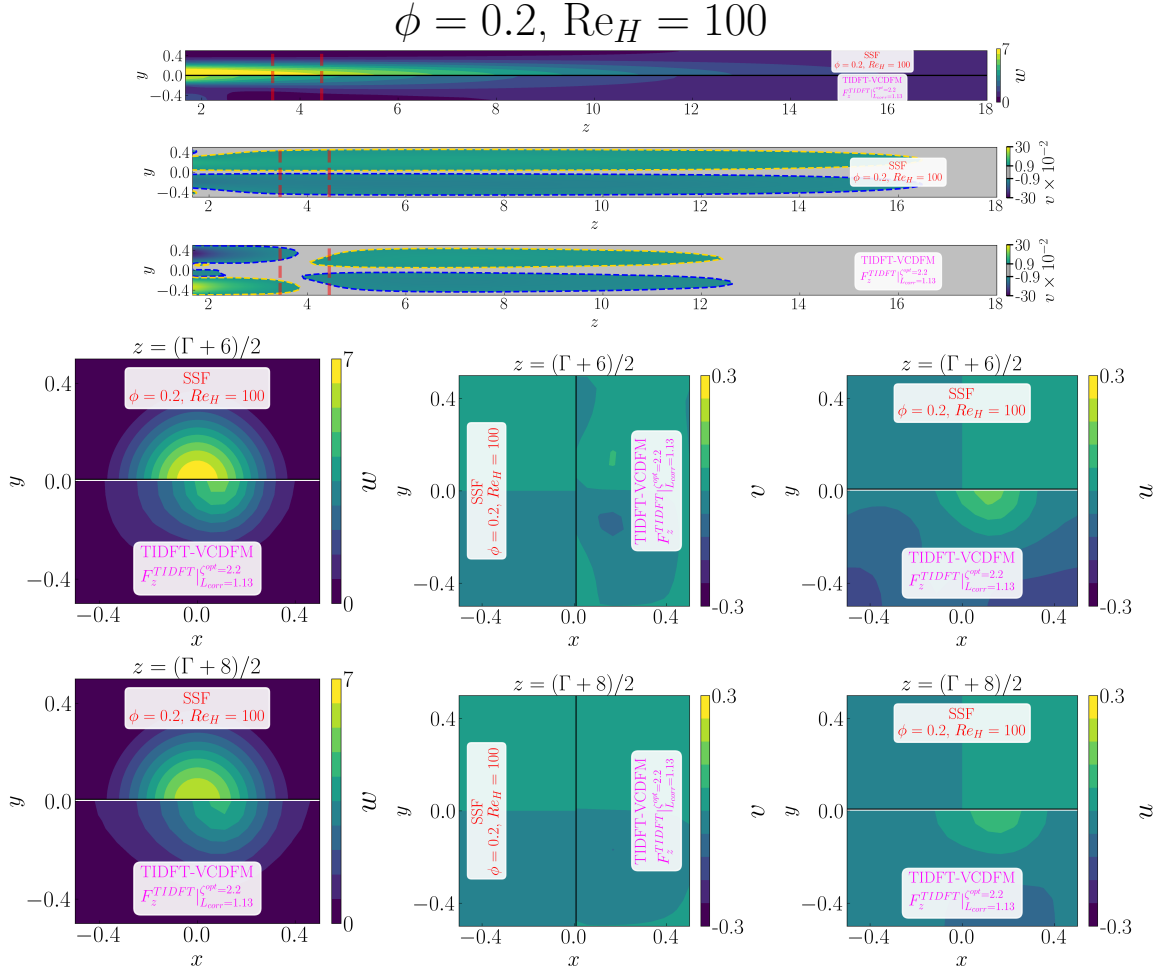


Figure D.250: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.2$ and $\text{Re}_H = 100$. The upper panels compare the downstream flow in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions at $z = (\Gamma + 6)/2$ and $z = (\Gamma + 8)/2$.

D.4 ML-VCDFM Validation under Reynolds-Number Extrapolation

The machine-learning model presented in this work was trained using data up to $\text{Re}_H = 50$. The Reynolds number $\text{Re}_H = 100$ therefore represents an extrapolation case lying outside the training database.

This section compares the geometry-resolved small-scale flow (SSF) and ML-VCDFM at $\text{Re}_H = 100$ for the porosities considered in this appendix. These comparisons assess the effect of extrapolating the spectral machine-learning model beyond its training Reynolds-number range. They show whether the extrapolated forcing preserves the downstream correction and whether the resulting ML-VCDFM reconstruction remains consistent with the SSF reference.

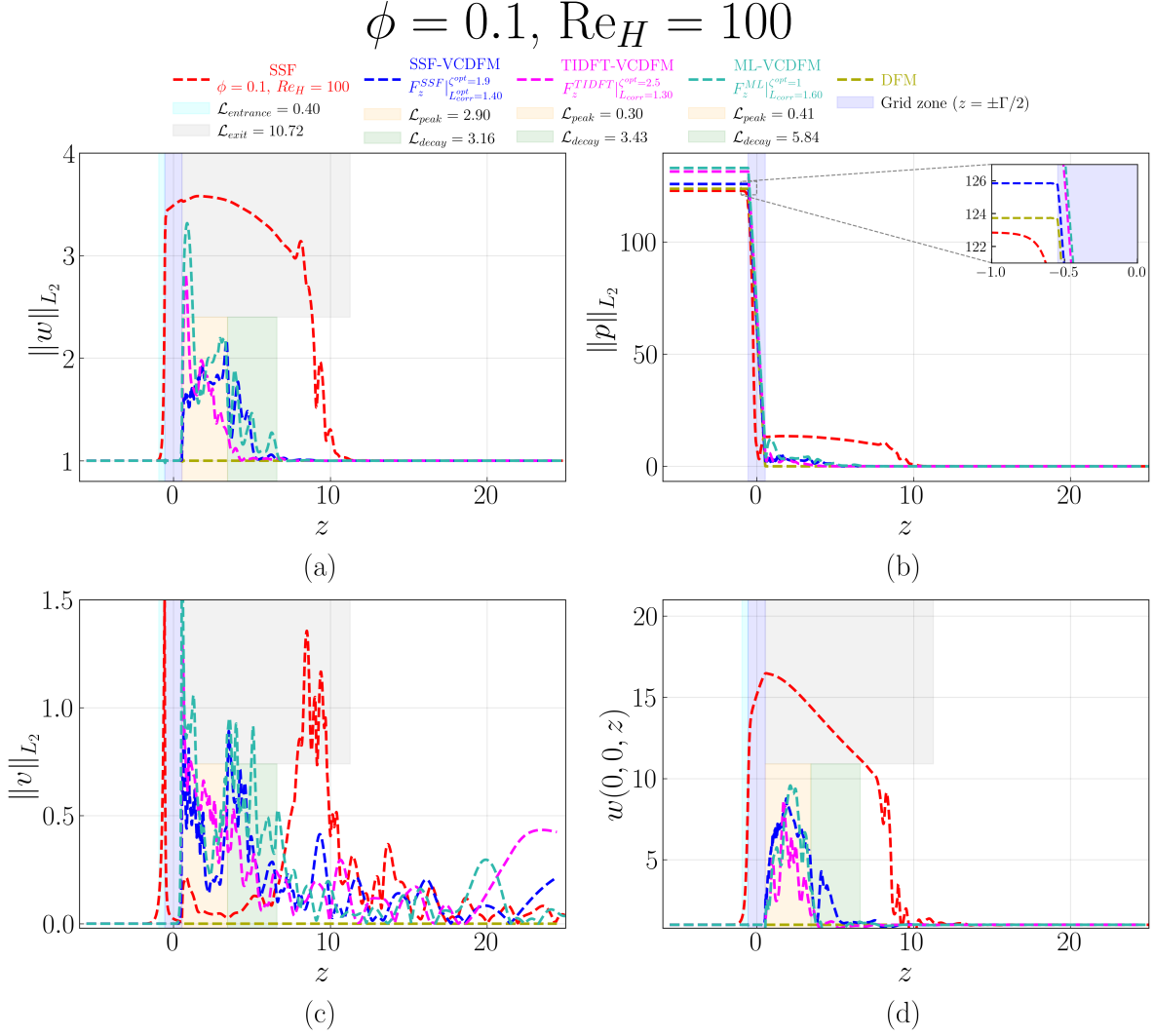


Figure D.251: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 100$. The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend. The ML-VCDFM result tests extrapolation beyond the Reynolds-number range used for training and deviates from the reference SSF solution.

$$\phi = 0.1, \text{Re}_H = 100$$

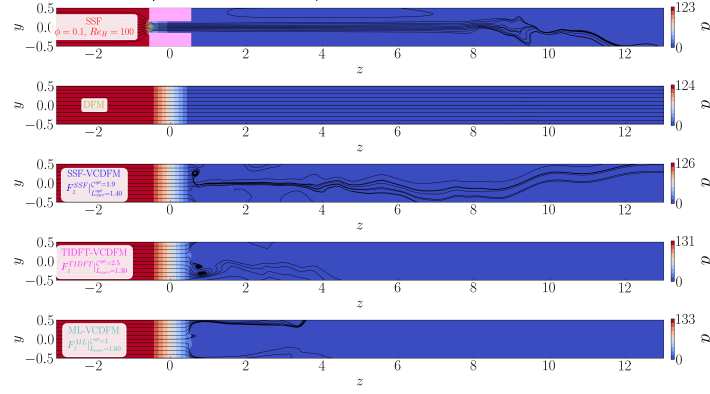
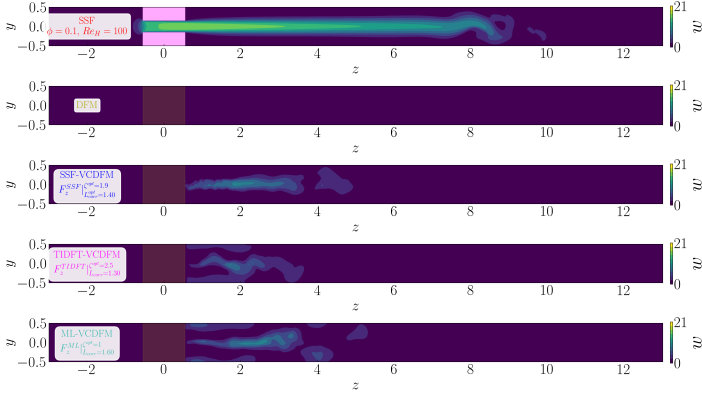
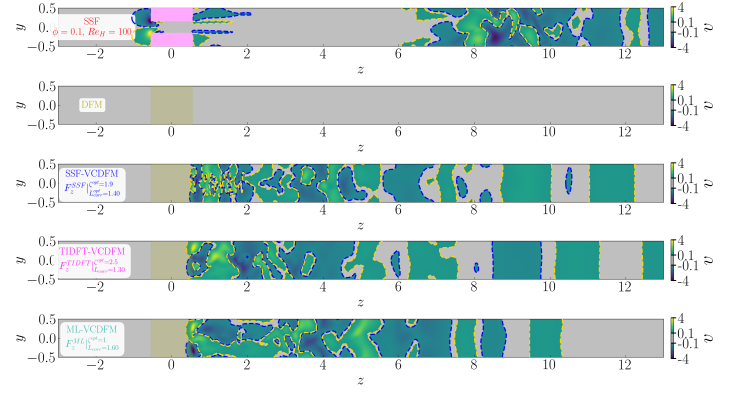

 (a) Pressure p .

 (b) Streamwise velocity w .

 (c) Transverse velocity v .

Figure D.252: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.1$, $\text{Re}_H = 100$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. The ML-VCDFM result tests extrapolation beyond the Reynolds-number range used for training and shows deviations from the reference SSF. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

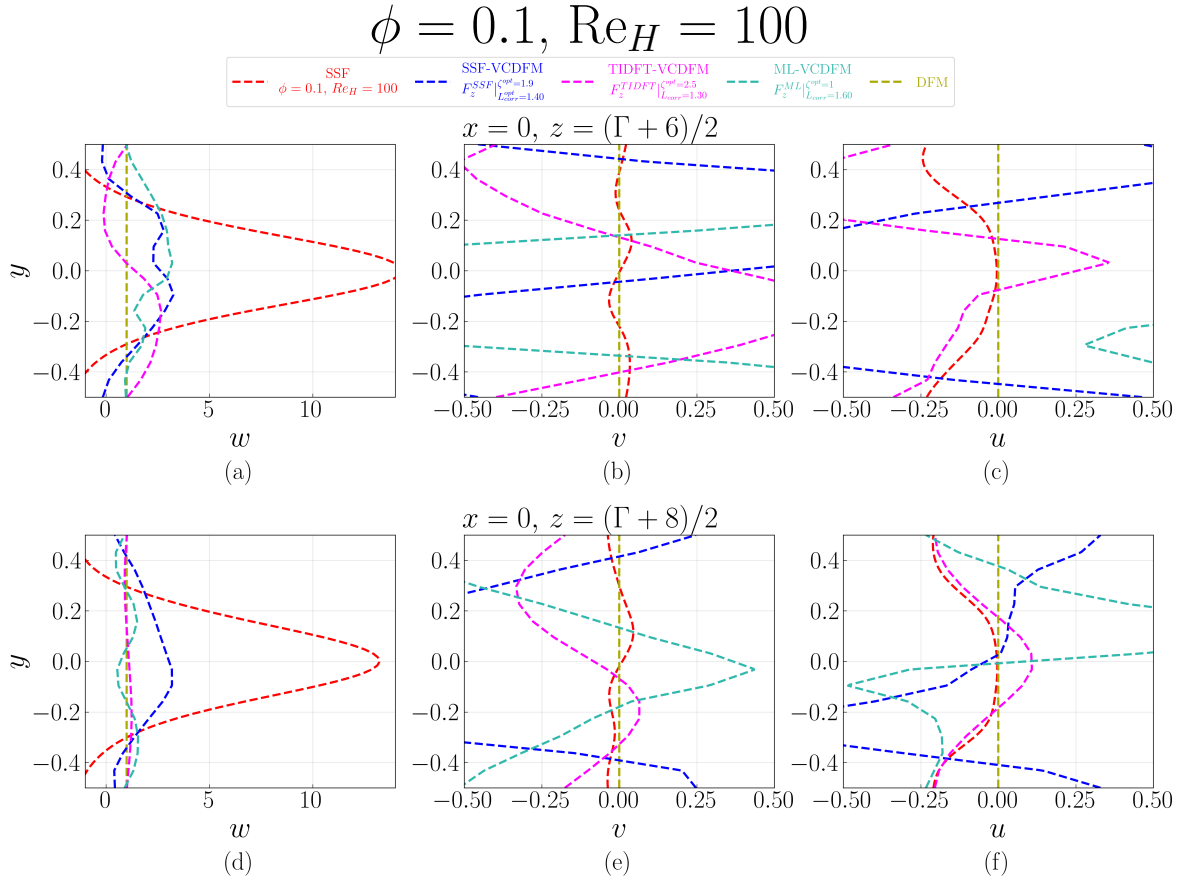


Figure D.253: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 100$, comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (\Gamma + 6)/2$ in (a–c) and $z = (\Gamma + 8)/2$ in (d–f). ML-VCDFM is evaluated outside the Reynolds-number range used for training.

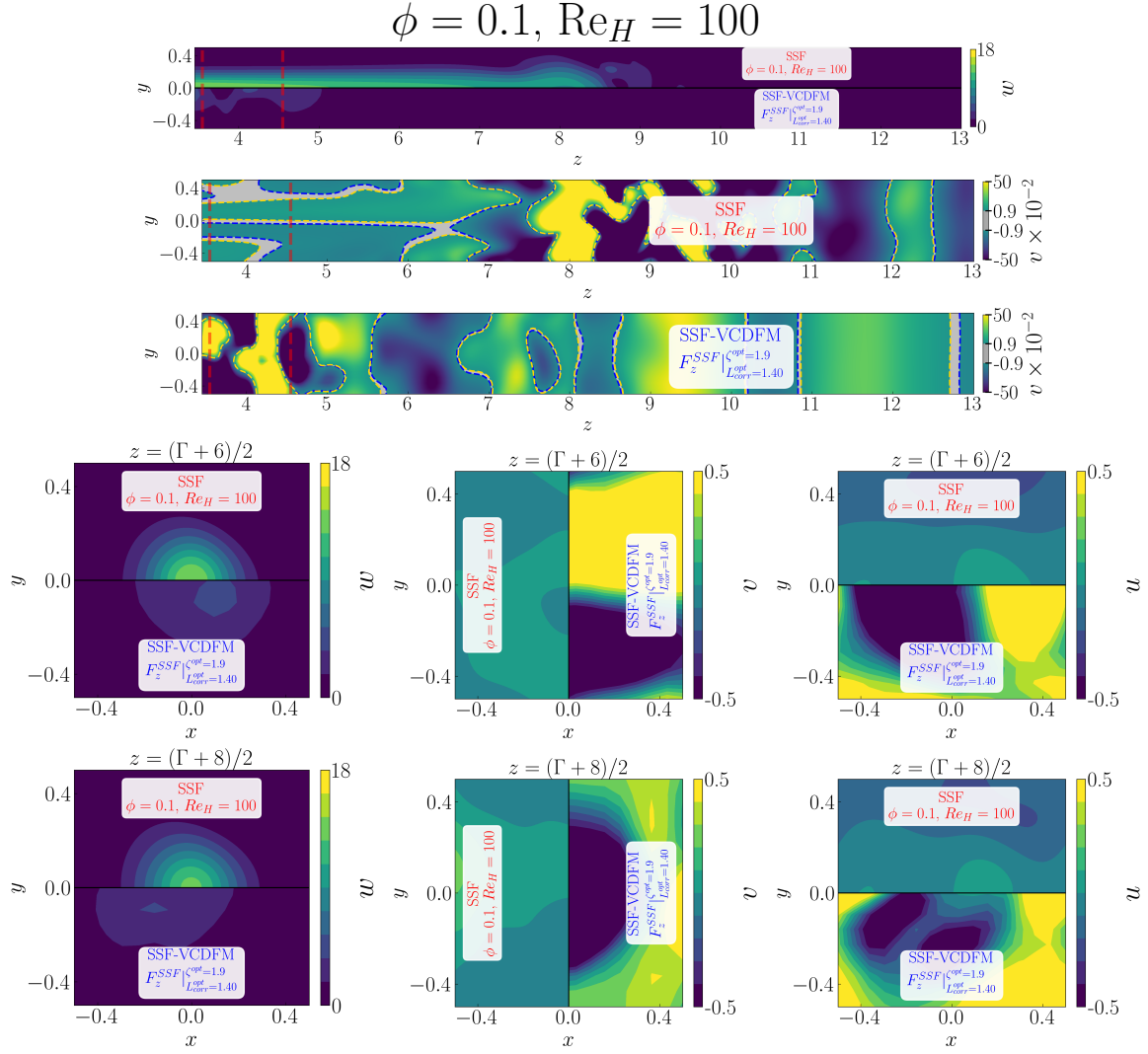


Figure D.254: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 100$. The upper panels compare streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 6)/2$ in the first row and $z = (\Gamma + 8)/2$ in the second row.

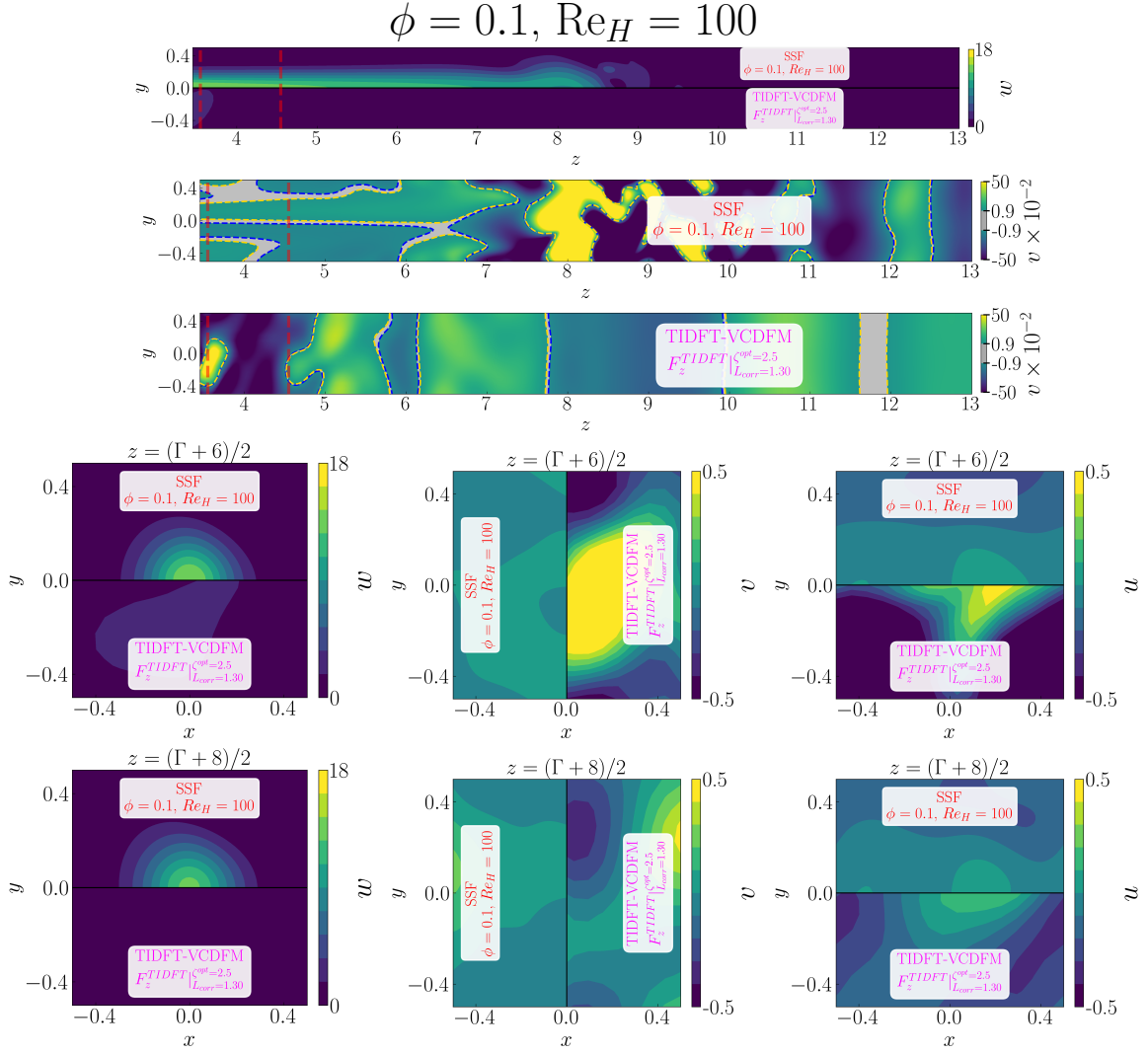


Figure D.255: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 100$. The upper panels compare streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 6)/2$ in the first row and $z = (\Gamma + 8)/2$ in the second row.

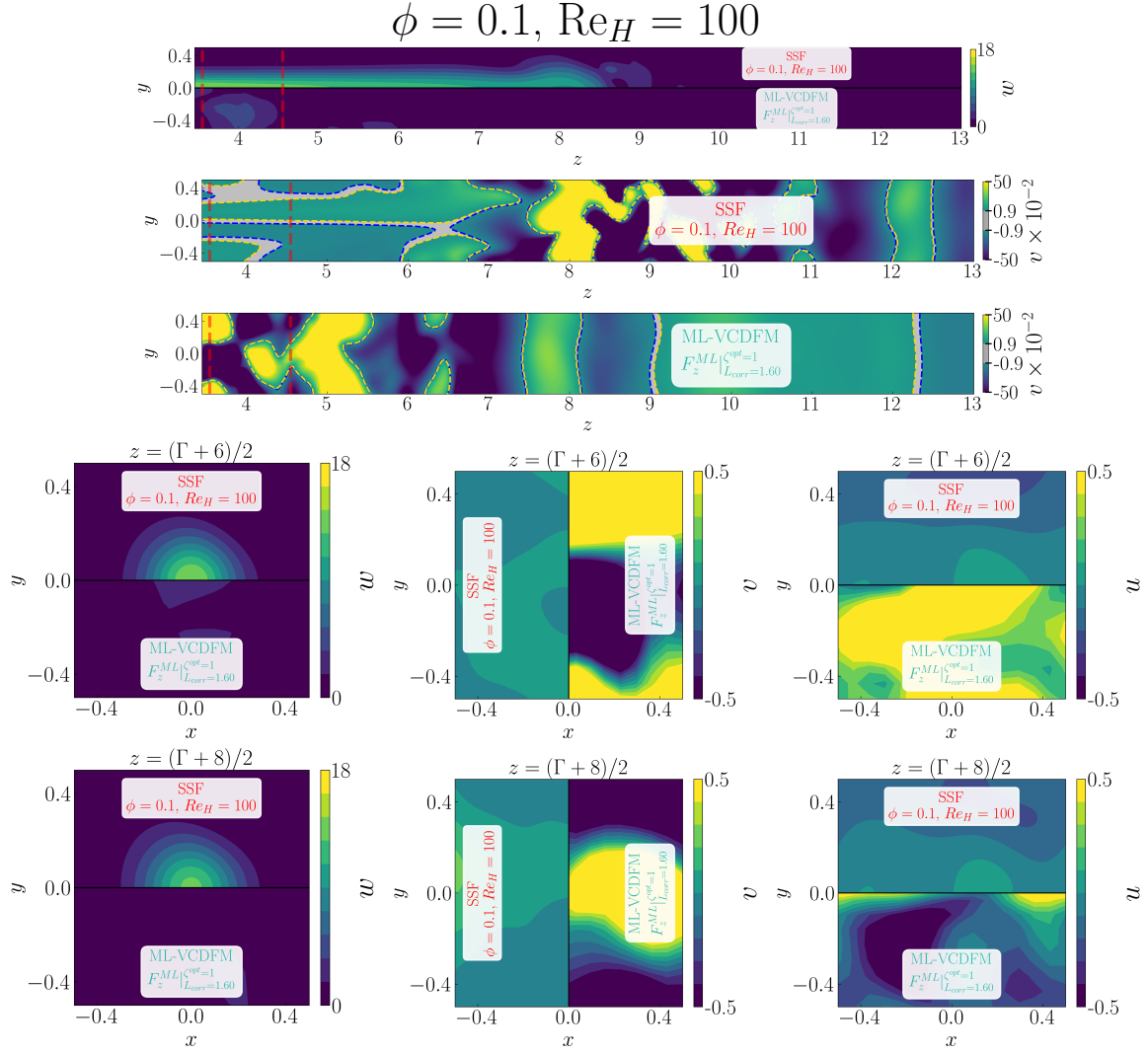


Figure D.256: Detailed comparison between the geometry-resolved small-scale flow (SSF) and ML-VCDFM using F_z^{ML} for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 100$. The upper panels compare streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 6)/2$ in the first row and $z = (\Gamma + 8)/2$ in the second row. This case tests extrapolation beyond the Reynolds-number range used for training; deviations from the SSF appear in the velocity fields.

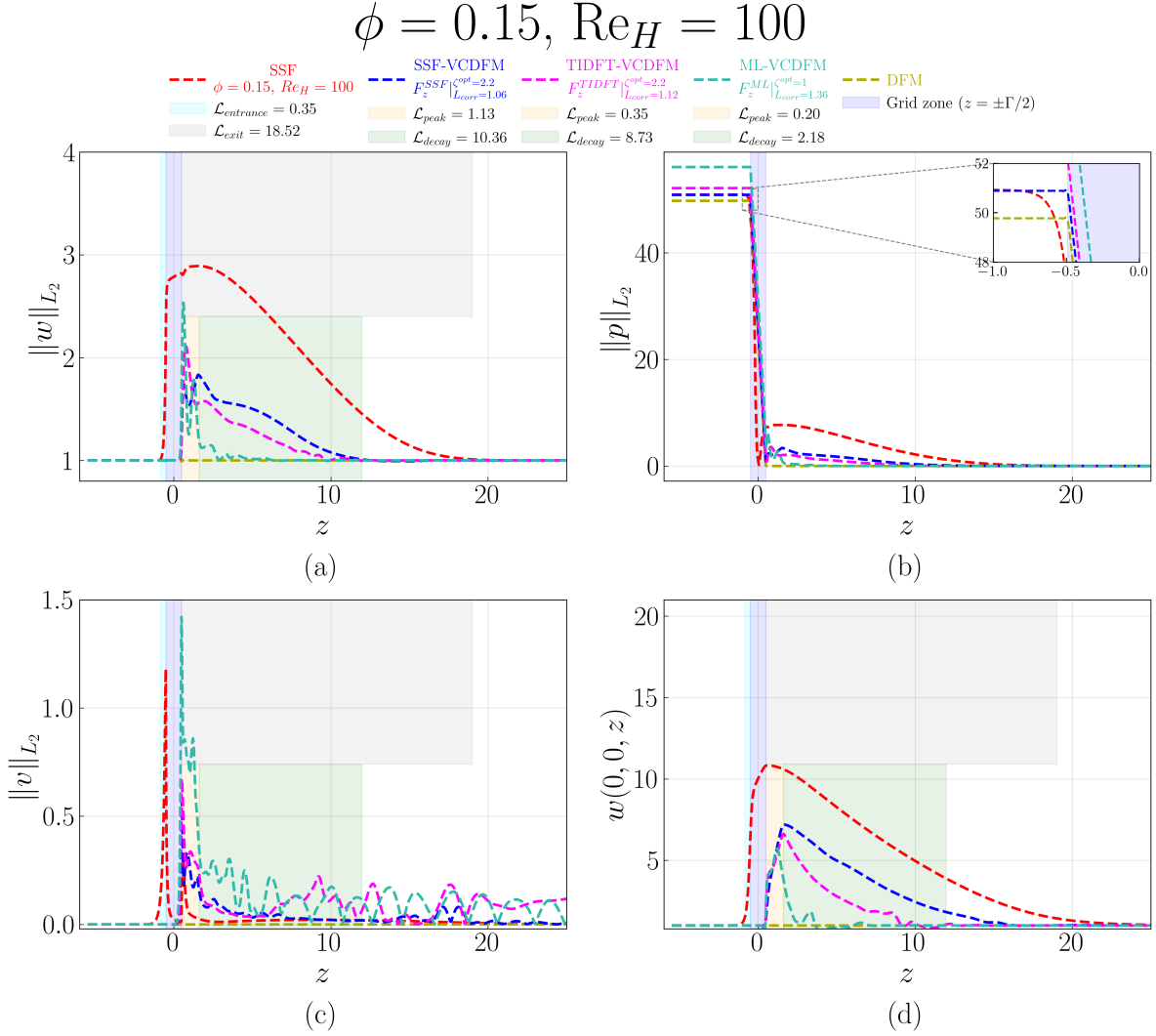


Figure D.257: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.15$ and $\text{Re}_H = 100$. The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend. The ML-VCDFM result tests extrapolation beyond the Reynolds-number range used for training and deviates from the reference SSF solution.

$$\phi = 0.15, \text{Re}_H = 100$$

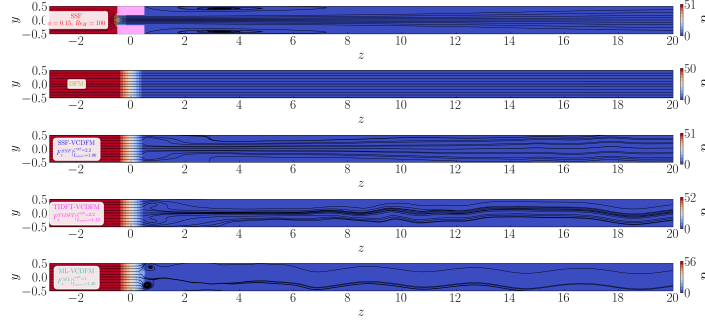
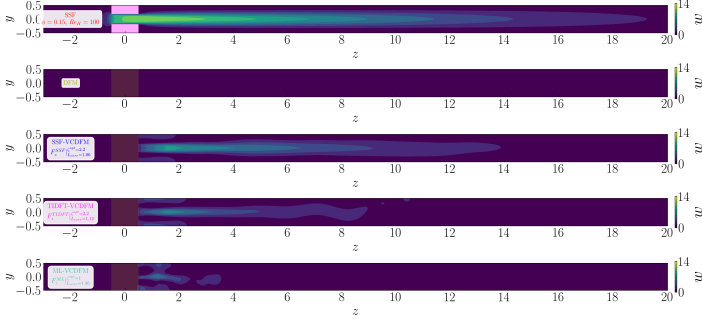
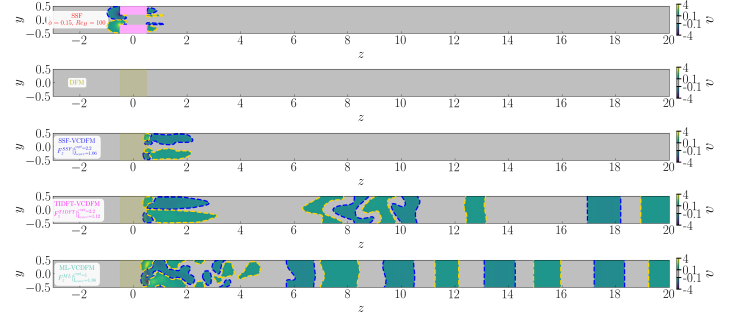

 (a) Pressure p .

 (b) Streamwise velocity w .

 (c) Transverse velocity v .

Figure D.258: Flow organization in the streamwise (z, y)-plane for the 1×1 SRG at $\phi = 0.15$, $\text{Re}_H = 100$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. The ML-VCDFM result tests extrapolation beyond the Reynolds-number range used for training and shows deviations from the reference SSF. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.